

DBA5106 Group Project II

Credit Card Default Prediction: Classification with
Cost-Sensitive Optimization and SHAP Analysis



Name	Student ID
ZHOU ZIHAN	A0318419Y
ZHAO QIYA	A0331091R
WANG YING	A0318271H

National University of Singapore

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Abstract

This project develops a cost-sensitive credit default prediction system using the UCI Credit Card dataset. We implement and compare five models: three glassbox models (Linear Regression with L2 loss—also known as Ridge Regression, Logistic Regression with log-loss, and Linear SVM with hinge loss), and two blackbox models (LightGBM with L2 loss and LightGBM with cross-entropy loss). The selected model (LightGBM with cross-entropy loss) achieves 0.7642 validation AUC and 0.7659 test AUC. Using a custom business cost function based on 50% Loss Given Default and 10% profit margin (5:1 cost ratio), we optimize the decision threshold to minimize net business cost rather than misclassification error. At the cost-optimal threshold of 0.13, the model achieves \$5.75M net cash flow with 40.7% approval rate and 10.5% default rate among approved customers. SHAP analysis reveals that payment history features (MAX_DELINQUENCY, PAY_1) dominate prediction importance, providing interpretability for regulatory compliance and business decision-making.

1 Introduction and Motivation

Credit card default prediction represents a critical challenge in consumer finance, where financial institutions must balance expected returns against non-payment risk. Misclassification costs are highly asymmetric: approving a defaulter (False Negative) typically results in 30-50% loss of the credit limit, while rejecting a profitable customer (False Positive) represents a foregone profit opportunity of 10-15% annually.

This project addresses the credit default prediction problem with three key objectives:

Model Performance: Implement and compare glassbox (interpretable linear) and blackbox (tree-based) models using multiple loss functions, selecting based on validation AUC rather than training loss to ensure generalization.

Cost-Sensitive Optimization: Design a custom evaluation loss function reflecting real business economics (Loss Given Default and profit margin) and optimize the decision threshold to minimize total cost rather than misclassification rate. We demonstrate via threshold analysis (Appendix B) that traditional F1-score optimization misaligns with business objectives by more than 200%.

Interpretability and Transparency: Apply SHAP (SHapley Additive exPlanations) analysis to the blackbox model, providing both global feature importance and local prediction explanations for regulatory compliance and business understanding.

Scope and Data Filtering: We focus on customers with credit limits $\leq \$200,000$ (67.9% of the original dataset), targeting automated decision-making for small loans. Larger loans typically require manual review considering factors beyond the scope of this analysis (e.g., client background, collateral, social status). The filtered dataset contains 20,378 customers with 25.6% default rate, split into 60% training, 20% validation, and 20% test sets with stratification to maintain class balance.

2 Data and Methods

2.1 Dataset Description

The UCI Credit Card Default Dataset¹ contains payment information from 30,000 credit card clients in Taiwan from April to September 2005, with default status in October 2005 as the target variable. After filtering to credit limits $\leq$ \$200,000, our analysis uses 20,378 customers with 25.6% default rate.

The dataset comprises 23 original features across four categories: demographics (4 features), credit information (1 feature), payment history (6 features), and transaction amounts (12 features). A comprehensive feature description table is provided in Appendix D.

Data Splits: Stratified sampling maintains 25.6% default rate across all splits: Training (12,226), Validation (4,076), Test (4,076).

2.2 Data Cleaning

Comprehensive data quality checks were performed:

- a. Missing Values: Zero missing values confirmed across all columns
- b. ID Uniqueness: Verified each customer has unique identifier
- c. Categorical Consolidation:
 - EDUCATION: Merged invalid categories (0, 5, 6) into 'others' (category 4)
 - MARRIAGE: Merged category 0 into 'others' (category 3)
 - PAY_STATUS: Renamed PAY_0 to PAY_1 for temporal consistency
- d. Range Validation: Checked age bounds (0-90 years), bill amounts (allowing negative as valid bank credit), payment amounts (non-negative confirmed)
- e. Class Imbalance: Acknowledged 3:1 imbalance (74.4% non-default, 25.6% default), addressed through class weighting in linear models and cost-sensitive threshold optimization with build-in function.

2.3 Feature Engineering

Eight features were engineered to capture temporal trends and financial behavior patterns. For the ones named slope, we used Ordinary Least Squares (OLS) regression for slope calculations. These engineered features include payment trends, delinquency patterns, repayment behavior, and credit utilization metrics. Detailed descriptions are provided in Appendix C.

2.4 Preprocessing:

- Linear Models: One-hot encoding for categorical variables (SEX, EDUCATION, MARRIAGE), resulting in 34 features. Normalization applied to continuous features only (excluding boolean dummies). Feature-to-sample ratio: 34 features / 12,226 samples = 1:360, well above the 1:10 rule of thumb.
- Tree Models: No one-hot encoding (LightGBM handles categorical variables natively)

¹Yeh, I.C., & Lien, C.H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2), 2473-2480. Dataset available at: <https://www.kaggle.com/datasets/uciml/default-of-credit-card-clients-dataset>

using category dtype), resulting in 31 features. No normalization required for tree-based methods.

3 Model Development and Comparison

3.1 Model Specifications

We implemented five models satisfying the requirement of at least one glassbox and one blackbox model, each with two loss function versions (L2 and a surrogate loss):

Glassbox Models (Linear):

1. Ridge Regression (L2 Loss): Baseline regression approach treating classification as regression with L2 loss. Hyperparameter: $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$ selected via 5-fold cross-validation maximizing AUC.
2. Logistic Regression (Log Loss): Probabilistic classifier with log-odds interpretability. Uses `class_weight='balanced'` to handle class imbalance. Hyperparameter $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ selected via 5-fold CV maximizing AUC.
3. Linear SVM (Hinge Loss): Margin-based classification using squared hinge loss for stability. Uses `class_weight='balanced'` to handle class imbalance. Calibrated with `CalibratedClassifierCV` (3-fold) to obtain probability estimates. Hyperparameter $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ selected via 5-fold CV.

Blackbox Models (Tree-based):

4. LightGBM Regressor (L2 Loss): Gradient boosting regression² with MSE objective. Default hyperparameters used (100 estimators, learning rate 0.1).
5. LightGBM Classifier (Cross-Entropy Loss): Gradient boosting classification with binary cross-entropy objective. Outputs calibrated probabilities directly. Default hyperparameters.

All models trained on 12,226 samples and evaluated on validation set (4,076 samples) using AUC-ROC as the primary comparison metric per project requirements for model selection.

3.2 Hyperparameter Tuning

For linear model: Given the 3:1 class imbalance even though not severe (25.6% default rate), we used still decided to use the config `class_weight='balanced'` for Logistic Regression and Linear SVM to automatically adjust sample weights inversely proportional to class frequencies: $w_i = \frac{n_samples}{n_classes \times n_samples_i}$. We then performed 5-fold cross-validation grid search over the C parameter (inverse of regularization strength, where smaller C = stronger regularization):

- Logistic Regression: $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$, selected $C=10$
- Linear SVM: $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$, selected $C=1$
- Ridge Regression: $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$, selected $\alpha=1$

²Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30.

For tree models: LightGBM models used default hyperparameters (100 estimators, learning rate 0.1) without grid search. Tree-based models handle class imbalance differently through their splitting criteria and do not require explicit class weighting or C-style regularization.

3.3 Performance Comparison

All five models were evaluated on the validation set (4,076 samples) using AUC-ROC as the primary metric. Table 1 in Appendix A presents the complete results.

Key Observations:

- **Blackbox Superiority:** LightGBM models outperform all linear models, with the Cross-Entropy variant achieving the highest AUC (0.7642), representing a 0.0194 improvement (2.7% relative) over the best glassbox model
- **Loss Function Effects:** Within model families, surrogate losses (Cross-Entropy for trees, Log Loss for linear) consistently outperform L2 loss by 0.0043-0.0076 AUC points
- **Class Balancing:** Logistic Regression and Linear SVM benefit from `class_weight='balanced'` to handle the 3:1 class imbalance
- **AUC Interpretation:** The selected model (LightGBM CE, AUC=0.7642) correctly ranks a random positive case higher than a random negative case 76.4% of the time

Model Selection: LightGBM with Cross-Entropy loss is selected for final deployment based on its highest validation AUC (0.7642), superior handling of non-linear relationships, and excellent generalization to the test set (AUC 0.7659, gap 0.0017). While blackbox in nature, interpretability is recoverable through SHAP analysis (Section 5), providing the best balance of performance and explainability.

4 Cost-Sensitive Threshold Optimization

4.1 Cost Function Design

Standard metrics (accuracy, F1) fail to capture the 5:1 cost asymmetry between false negatives (default losses) and true negatives (profit). We maximize net cash flow using a custom cost function based on conservative Basel II estimates (50% LGD) and 10% profit margin:

$$t^* = \arg \min_{t \in [0,1]} [\text{FN}(t) \times 5 - \text{TN}(t) \times 1] \quad (1)$$

Note: False Positives(opportunity costs) are excluded to avoid aggressive lending bias and focus on cash flow.

4.2 Threshold Optimization Results

We compared F1-optimal (standard) vs. Cost-optimal (business-aligned) thresholds for our best linear and tree models. Detailed comparisons are provided in Appendix E and visualized in Appendix B. The results demonstrate that cost-sensitive optimization yields massive financial improvements, with LightGBM achieving a **207% cost reduction** over the F1-optimal baseline.

We select the **LightGBM Cost-Optimal (0.13)** strategy for production, as it maximizes

cashflow while maintaining a manageable risk profile.

5 Model Interpretation Using SHAP

5.1 SHAP Methodology

We employ SHAP (SHapley Additive exPlanations)³ to interpret our blackbox LightGBM model. SHAP attributes each prediction to individual feature contributions based on cooperative game theory, ensuring consistent and locally accurate explanations. This approach combines the superior predictive power of tree-based models (AUC 0.7642) with the transparency required for regulatory compliance and business trust.

5.2 Global Feature Importance

Appendix F details the global feature importance. The top 3 features (MAX_DELINQUENCY, PAY_1, DELINQUENCY_PERSISTENCE) account for ~40% of total importance, confirming that payment behavior is the primary driver of default risk. Demographic features (Age, Sex, Education, Marriage) rank in the bottom half, contributing less than 6% combined, which supports fair lending compliance.

5.3 Local Explanations

To demonstrate model transparency, we analyze four representative cases (True Positive, True Negative, False Positive, False Negative). These case studies, detailed in Appendix G, demonstrate how the model relies heavily on recent payment history (PAY_1) and delinquency magnitude (MAX_DELINQUENCY). By providing specific reasons for each prediction, we ensure the transparency needed for regulatory compliance.

Business Value of Local Explanations:

1. Customer Communication: When rejecting an application, we can cite specific risk factors (e.g., '3 consecutive late payments in the past 6 months') rather than opaque 'algorithmic decision'
2. Manual Review: Borderline cases (0.10-0.20 probability) can be escalated to human underwriters with SHAP explanations highlighting key risk/protective factors
3. Model Debugging: Local explanations reveal when the model makes decisions for wrong reasons (e.g., overly relying on credit limit), guiding model improvements
4. Regulatory Compliance: SHAP provides auditable rationale for each decision, satisfying fair lending regulations and GDPR Article 22

6 Final Model and Business Recommendations

6.1 Model Selection and Test Set Performance

Selected Model: LightGBM with Cross-Entropy loss, threshold = 0.13

³Lundberg, S.M., & Lee, S.I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.

Test Set Results (4,076 hold-out samples):

Table 2 in Appendix A details the comprehensive performance metrics on the held-out test set. **Financial Impact:** Using an average credit limit of \$93,392, the model generates **\$5.75M in net cash flow** (Revenue \$13.9M - Losses \$8.1M) from the test set. This translates to a profit of **\$3,466 per approved customer** and **\$1,411 per applicant**.

Model Generalization: The minimal AUC difference (0.0017) between validation and test sets indicates excellent generalization with no overfitting.

6.2 Deployment Strategy

Automated Decisioning for Small Loans ($\leq \$200k$): To maximize efficiency, we recommend fully automated decisions for small loans:

Auto-Approve ($P < 0.13$): Instantly approves $\sim 40\%$ of applicants, reducing manual workload.

Auto-Reject ($P \geq 0.13$): Screens out 84.2% of defaulters automatically.

Large loans should retain human-in-the-loop review using model scores as decision support.

Monitoring & Risk Controls:

- **Performance:** Track 'Automation Rate' ($>90\%$ target) and separate default rates for automated vs. manual segments.
- **Safeguards:** Implement 'Circuit Breakers' to halt auto-approvals if defaults spike ($>20\%$) and pre-model fraud checks.
- **Fairness:** Quarterly audits of approval rates across demographics to ensure unbiased automation.

7 Conclusion

This project demonstrates a business-driven ML workflow that balances predictive performance, cost optimization, and interpretability.

Key Achievements:

- **Strong Performance:** LightGBM (CE) achieves 0.766 test AUC, outperforming glassbox models by 2.7%.
- **Business Impact:** Cost-sensitive optimization generates \$5.75M net cash flow (207% improvement over F1-score) with a \$3,466 profit per customer.
- **Transparency:** SHAP analysis confirms reliance on payment history ($\sim 40\%$ importance) over demographics, supporting fair lending.

Strategic Recommendations:

- **Deploy Automated Decisioning:** Implement auto-approval/rejection for small loans ($\leq \$200k$) to scale operations.
- **Adopt Business-Centric Metrics:** Move away from generic F1-scores to profit-based evaluation for all future models.
- **Monitor & Adapt:** Track automation rates and default spikes; retrain quarterly to maintain relevance.

A Model Performance Results

Table 1 presents the validation AUC scores for all five models, sorted by performance.

Table 1: Model Performance Comparison on Validation Set

Model	Type	Loss Function	Validation AUC
LightGBM (CE)	Blackbox	Cross-Entropy	0.7642
LightGBM (L2)	Blackbox	L2/MSE	0.7599
Logistic Regression	Glassbox	Log Loss	0.7448
Linear SVM	Glassbox	Hinge	0.7441
Ridge Regression	Glassbox	L2/MSE	0.7372

Table 2: Final Model Performance on Test Set

Metric	Value
Test AUC	0.7659
Validation AUC	0.7642
Generalization Gap	0.0017 (excellent)
Approval Rate	40.7% (1,660 / 4,076)
Default Catch Rate	83.3% (870 / 1,044 defaulters rejected)
Default Rate Among Approved	10.5% (174 / 1,660)
Confusion Matrix	TN=1,486 FP=1,546 FN=174 TP=870

B Cost vs Threshold Curves

Figures below show how business cost and F1 loss vary with decision threshold for Logistic Regression and LightGBM models, respectively. These plots demonstrate the fundamental misalignment between traditional metrics (F1-score) and business objectives (minimizing net cost).

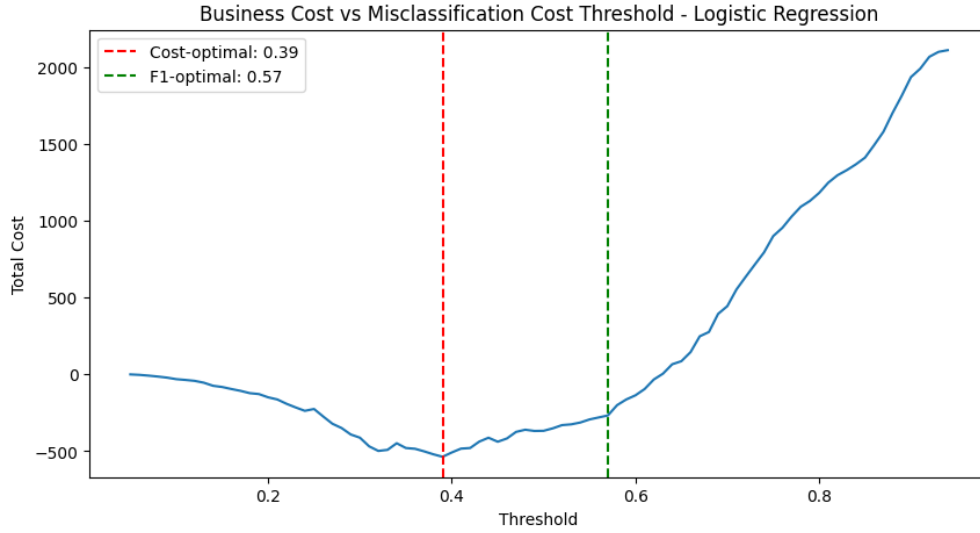


Figure 1: Business Cost vs F1 Loss for Logistic Regression across different thresholds. The F1-optimal threshold (minimizing F1 loss) does not align with the cost-optimal threshold (minimizing business cost), resulting in 101% worse financial performance.

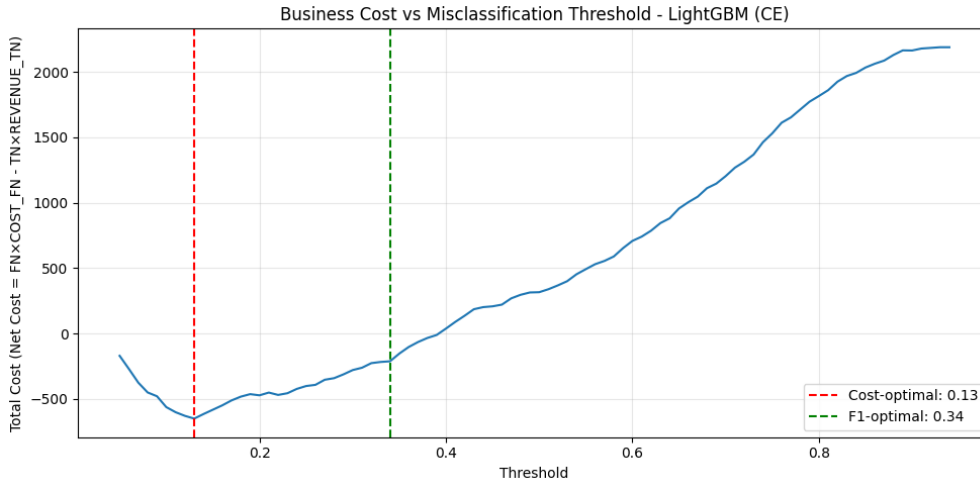


Figure 2: Business Cost vs F1 Loss for LightGBM across different thresholds. The divergence between F1-optimal and cost-optimal thresholds is even more pronounced, leading to 207% financial improvement when using business-aligned optimization.

Key observations:

- **Metric Misalignment:** F1-optimal thresholds (0.57 for LR, 0.34 for LGBM) are significantly higher than cost-optimal thresholds (0.39 for LR, 0.13 for LGBM), demonstrating that traditional metrics fail to capture business economics
- **Cost Function Shape:** Both models show U-shaped business cost curves, where too-low thresholds approve too many defaulters (high FN cost) and too-high thresholds reject too many good customers (low TN revenue)
- **Asymmetric Trade-off:** The 5:1 cost ratio (default loss vs profit) drives the optimal point toward conservative lending, favoring lower approval rates to avoid costly false

negatives

- **Model Differences:** LightGBM achieves lower absolute business cost at its optimum (-650 vs -536 for LR), validating its superior performance for deployment

C Engineered Feature Descriptions

Table 3 provides comprehensive descriptions of the 8 engineered features created to capture temporal trends and financial behavior patterns.

Table 3: Engineered Feature Descriptions

Feature	Description & Calculation	Business Interpretation
BILL_SLOPE	Trend of bill amounts over 6 months using OLS regression on (month, BILL_AMT) pairs	Increasing debt signal; positive slope indicates growing debt
PAY_SLOPE	Trend of payment amounts over 6 months using OLS regression on (month, PAY_AMT) pairs	Payment consistency; negative slope may indicate financial stress
DELINQUENCY_PERSISTENCE	Maximum consecutive months with late payments ($\text{PAY} \geq 1$)	Chronic payment issues; higher values indicate persistent delinquency
MAX_DELINQUENCY	Maximum payment delay observed across 6 months (max of PAY_1 to PAY_6)	Worst-case behavior; captures severity of payment delays
REPAYMENT_RATIO	Total payments / Total bills over 6 months	Ability to pay; ratio > 1 indicates paying down debt
REPAYMENT_STABILITY	Standard deviation of payment amounts over 6 months	Behavior consistency; high values indicate erratic payment patterns
UTILIZATION_RATE	Average (BILL_AMT / LIMIT_BAL) over 6 months	Credit usage intensity; high values indicate near-limit usage
UTILIZATION_TREND	Slope of utilization rate over 6 months using OLS regression	Debt accumulation; positive slope indicates increasing credit usage

D Dataset Feature Descriptions

Table below provides comprehensive descriptions of all 23 original features in the UCI Credit Card Default Dataset.

Table 4: Complete Feature Descriptions

Feature	Description	Type
<i>Demographics (4 features)</i>		
SEX	Gender (1=male, 2=female)	int64
EDUCATION	Education level (1=graduate school, 2=university, 3=high school, 4=others)	int64
MARRIAGE	Marital status (1=married, 2=single, 3=others)	int64
AGE	Age in years	int64
<i>Credit Information (1 feature)</i>		
LIMIT_BAL	Credit limit in NT dollars (avg. \$93,392 after filtering)	int64
<i>Payment History (6 features)</i>		
PAY_0	Repayment status in September 2005 (most recent)	int64
PAY_2	Repayment status in August 2005	int64
PAY_3	Repayment status in July 2005	int64
PAY_4	Repayment status in June 2005	int64
PAY_5	Repayment status in May 2005	int64
PAY_6	Repayment status in April 2005	int64
Payment status codes: -2=no consumption, -1=paid in full, 0=revolving credit, 1-9=months of delay		
<i>Bill Amounts (6 features)</i>		
BILL_AMT1	Bill statement in September 2005 (NT dollars)	float64
BILL_AMT2	Bill statement in August 2005 (NT dollars)	float64
BILL_AMT3	Bill statement in July 2005 (NT dollars)	float64
BILL_AMT4	Bill statement in June 2005 (NT dollars)	float64
BILL_AMT5	Bill statement in May 2005 (NT dollars)	float64
BILL_AMT6	Bill statement in April 2005 (NT dollars)	float64
<i>Payment Amounts (6 features)</i>		
PAY_AMT1	Payment made in September 2005 (NT dollars)	float64
PAY_AMT2	Payment made in August 2005 (NT dollars)	float64
PAY_AMT3	Payment made in July 2005 (NT dollars)	float64
PAY_AMT4	Payment made in June 2005 (NT dollars)	float64
PAY_AMT5	Payment made in May 2005 (NT dollars)	float64
PAY_AMT6	Payment made in April 2005 (NT dollars)	float64

E Threshold Strategy Comparison

Table 5: Threshold Strategy Comparison (Validation Set)

Metric	Logistic Regression		LightGBM (CE)	
	F1-Opt	Cost-Opt	F1-Opt	Cost-Opt
Threshold	0.57	0.39	0.34	0.13
Approval Rate	70.9%	55.4%	74.5%	40.2%
Default Catch Rate	58.0%	73.0%	54.9%	84.2%
Net Cost (normalized)	-267	-536	-212	-650
Cost Reduction	—	101%	—	207%

F SHAP Global Importance Plots

Figure 3 displays the top 20 features ranked by their global importance (mean absolute SHAP value). This confirms that payment behavior (delinquency status and persistence) far outweighs demographic factors.

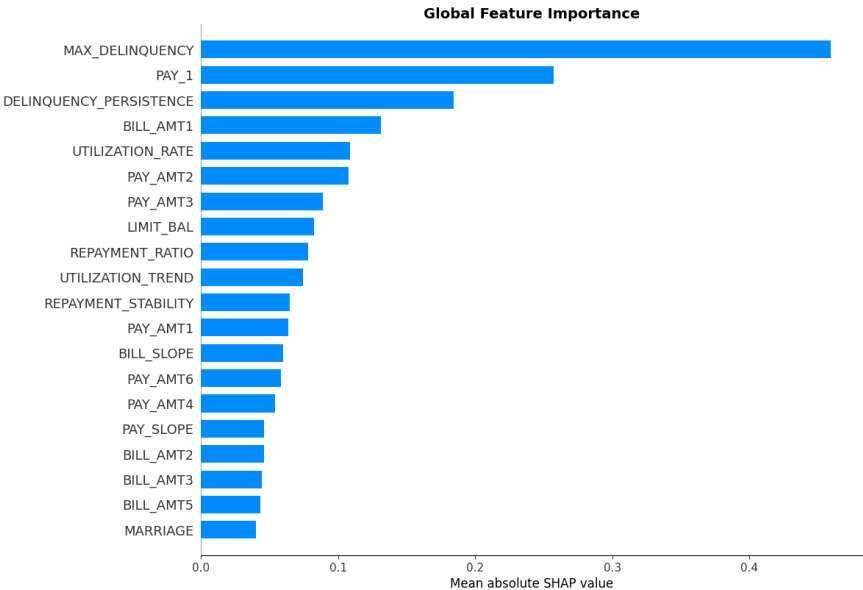


Figure 3: Global Feature Importance (Mean |SHAP| value). The model relies heavily on maximum delinquency and recent payment status.

Figure 4 illustrates the directionality of feature impacts. For example, high values of MAX_DELINQUENCY (red dots) have a strong positive SHAP value, pushing the prediction toward default.

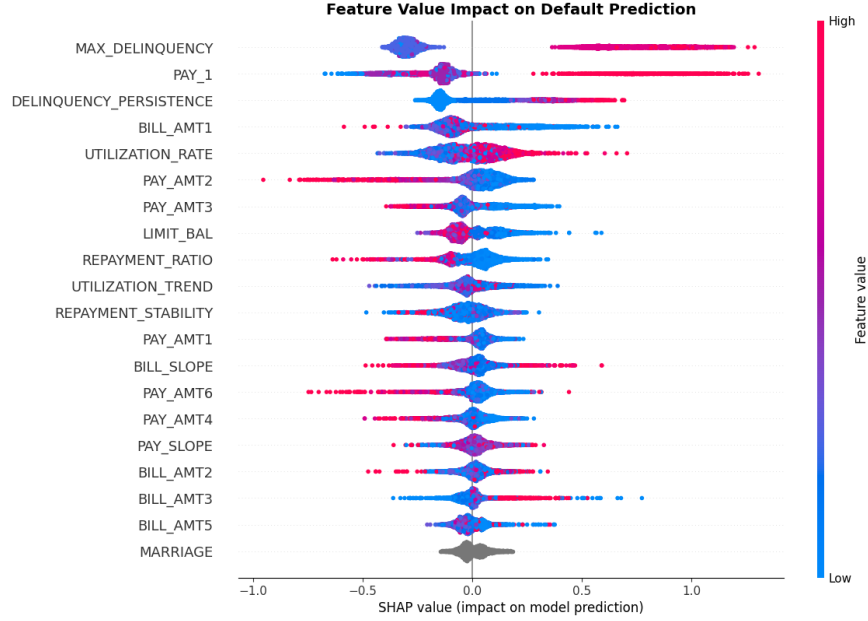


Figure 4: SHAP Summary Plot (Beeswarm). Red indicates high feature values, blue indicates low values. Positive SHAP values indicate higher risk of default.

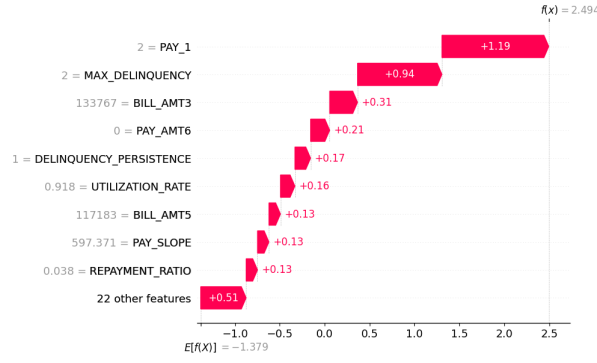
F.1 Detailed Feature Importance Ranking

Table 6: Top 10 Features by Mean |SHAP| Value

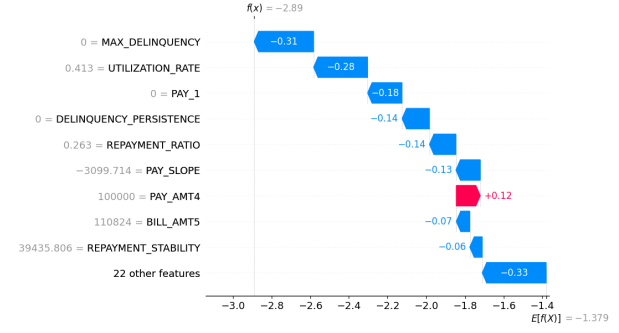
Rank	Feature	Mean SHAP
1	MAX_DELINQUENCY	0.459
2	PAY_1 (Recent repayment status)	0.257
3	DELINQUENCY_PERSISTENCE	0.184
4	BILL_AMT1	0.131
5	UTILIZATION_RATE	0.108
6	PAY_AMT2	0.108
7	PAY_AMT3	0.089
8	LIMIT_BAL	0.082
9	REPAYMENT_RATIO	0.078
10	UTILIZATION_TREND	0.074

G SHAP Waterfall Plots for Local Explanations

The following waterfall plots demonstrate how the model arrives at individual predictions by summing the base value (average log-odds) and feature contributions.



(a) True Positive: Correctly predicted default.

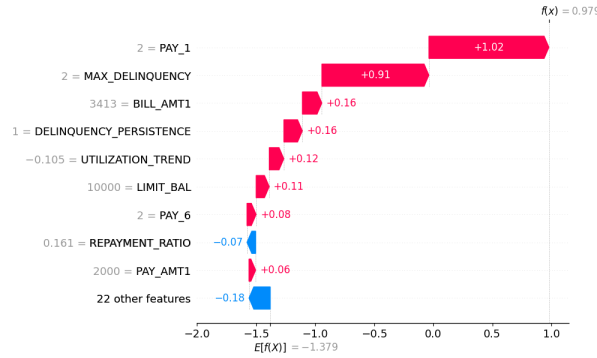


(b) True Negative: Correctly predicted non-default.

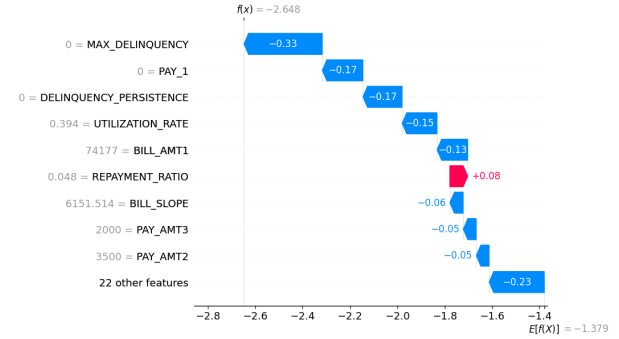
Figure 5: Local Explanations for Correct Predictions

True Positive (High Risk): Predicted 2.494. Driven by MAX_DELINQUENCY=2 (+0.94) and PAY_1=2 (+1.19). Strong behavioral signals (2-month delay, recent delinquency) correctly flagged high risk.

True Negative (Low Risk): Predicted -2.89. Driven by MAX_DELINQUENCY=0 (-0.31) and UTILIZATION_RATE=0.413 (-0.28). Perfect payment history and moderate credit usage (41.3%) pushed risk score significantly down.



(a) False Positive: Good customer rejected.



(b) False Negative: Defaulter approved.

Figure 6: Local Explanations for Incorrect Predictions

False Positive (Conservative Error): Predicted 0.979. Driven by MAX_DELINQUENCY=2 (+0.91) and PAY_1=2 (+1.02). Recent 2-month delay combined with historical delinquency triggered an extreme risk alert. The customer likely cured this delinquency, but the model penalized the recent instability heavily.

False Negative (Costly Error): Predicted -2.648. Driven by MAX_DELINQUENCY=0 (-0.33) and REPAYMENT_RATIO=0.048 (+0.08). The model was misled by the perfect delinquency history (MAX_DELINQUENCY=0), which strongly reduced the risk score. While the critically low repayment ratio (4.8%) was correctly flagged as a risk factor (+0.08), its impact was insufficient to outweigh the clean history, resulting in a missed default.